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Final Paper

From heating caves with burning wood to grinding grain with water-powered mills to the invention of the steam engine and electric grids, human innovation has rapidly expanded quality of life by harnessing and leveraging energy to perform work. In the first quarter of the twenty-first century, we face critical opportunities to revolutionize global energy systems, addressing two primary challenges: creating resilient, equitable energy distribution and preventing irreversible damage from global warming.

Currently, almost 10% of the world's population lacks reliable electricity, and developed countries' electrical grids require updates to increase capacity resiliently (Saundry, Electric Grid Overview, 2024). Energy consumption per capita strongly correlates with standard of living, making equitable energy access essential. However, most global energy still derives from fossil fuels, releasing substantial greenhouse gases (Andrews & Jelley, 2022, p. 439). Expanding energy systems could improve humanity's condition, but if done in a way that accelerates climate change, the consequences would be severe: billions in damages, mass deaths from extreme weather and droughts, ecosystem extinctions, and political instability (Andrews & Jelley, 2022, p. 439-441). The current goal set by the Paris Agreement in 2015 is to try to keep global warming below 2°C above pre-industrial levels, and ideally below 1.5°C (Saundry, Climate Change Science, 2024). Facing these challenges will require technological advancements, infrastructure changes, and global policy and cooperation.

While the politics of decarbonization are rather volatile, in the realm of technology, there are ample opportunities to decarbonize the world's energy system and potential for future innovation to overcome today's challenges. There is no single straightforward, one-size fits all solution. The ultimate decarbonization process will likely involve a broad range of technologies implemented on different timelines in different countries. However, it will almost certainly involve a phasing out of fossil fuel energies in favor of low-carbon generation technologies, the electrification of processes that currently use combustion, and the advancement of infrastructure to improve the efficiency, resiliency, and transportation of energy.

Energy is classified as either primary energy, derived directly from natural resources like fuels and renewables, or secondary energy, created by transforming a primary energy source such as electricity (Saundry, Energy Basics, 2024). Fuels generally produce heat energy which can be used directly, converted into mechanical energy for transportation, or transformed into mechanical energy to spin a generator which produces electrical energy

(Saundry, Energy Basics, 2024). Energy sources such as wind, solar, and hydraulic energy can be directly transformed into electric energy, thus reducing the amount of energy lost in conversion (Saundry, Energy Basics, 2024). Decarbonization can therefore largely be split into the decarbonization of primary energy sources and the decarbonization of secondary energy sources, mainly electricity. This focus is critical, as electricity generation accounts for a third of energy use in the United States, and 61% of global electricity still comes from fossil fuels (Saundry, Electric Grid Overview, 2024). Therefore, decarbonizing electricity and subsequently electrifying fuel-dependent processes is essential to decarbonizing the world's energy system.

From early human history until the industrial revolution, energy primarily came from burning of traditional biofuels, which are still used in much of the world today for heating and cooking (Saundry, Biofuels in Context, 2024). After the onset of the industrial revolution, coal, oil, and gas became ubiquitous both as a fuel source and eventually for electricity generation. In the 20th century, hydropower and nuclear power joined coal burning plants as the primary sources of electric generationand remained dominant through the end of the 20th century as seen in Figure 1 (IEA, 2024). All these established forms of energy face challenges as the energy landscape changes entering the 21st century, and all play a role in the decarbonization of the electric grid.

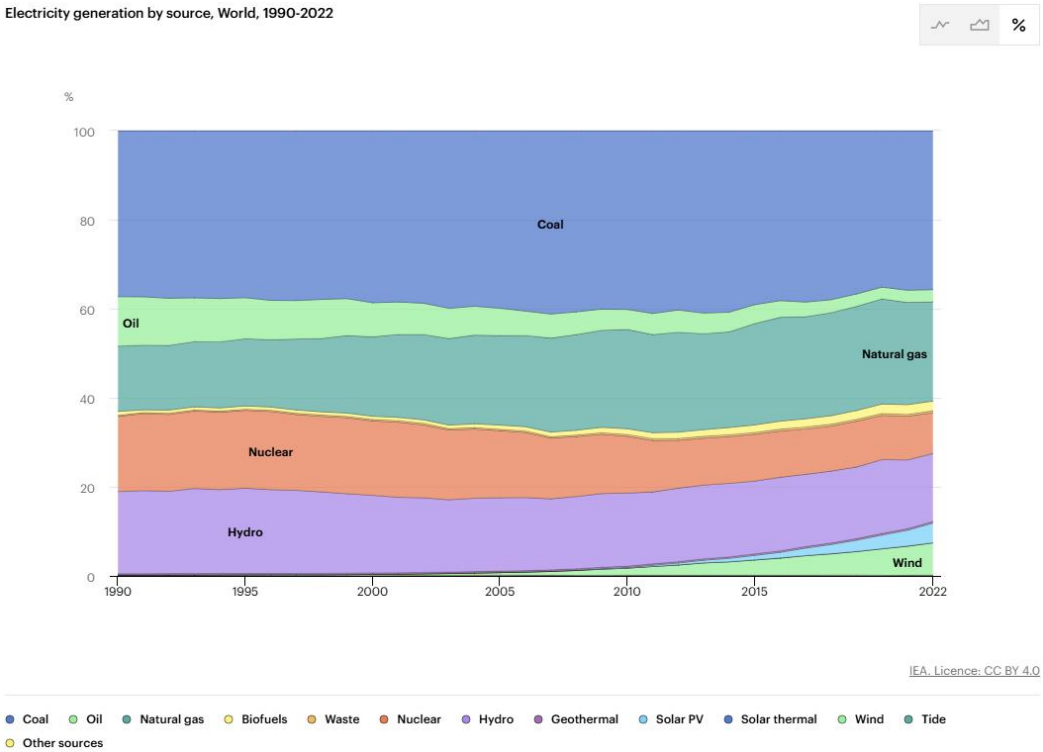

Figure 1

Fossil fuels are the most deeply embedded energy source in the global energy system today with coal still accounting for 35% of global electricity generation in 2022 (Saundry, Coal in Context, 2024). However, while energy infrastructure is still largely optimized for coal, the inefficiency of coal power plants and the levelized costs of electricity (LCOE) are making them increasingly uncompetitive in the 21st century, with 42% of coal-fired capacity unprofitable in 2018 (Andrews & Jelley, 2022, p. 115). Power plants that use fossil fuels must pay for the primary energy to convert it into electricity, but only 32% of the energy in the coal actually ends up being usable electricity; most is lost as heat (Saundry, Heat Rate, 2024). Coal has the highest carbon intensity of all fossil fuels with approximately 1000 kg CO_2eq MWh_e^{-1}, making it particularly important to reduce or eliminate coal-fired power generation in order to decarbonize the grid (Andrews & Jelley, 2022, p. 86).

While coal usage is decreasing in the US and decelerating in places like India and China, natural gas use has risen dramatically in the 21st century, particularly in the United States (Saundry, Heat Rate, 2024). This transition from coal to natural gas has both benefits and drawbacks. Natural gas generation, particularly combined cycle gas turbines, jas efficiencies of up to 64% which is nearly double the efficiency of coal powerplants (Andrdews & Jelley, 2022, p. 109). The fracking boom in the US has made natural gas relatively cost competitive. Due to its high efficiency and low carbon intensity compared to coal, this has helped drive a decrease in CO_2 from electricity emissions in the US by about 40% since 2005 (Saundry, Climate and Energy, 2024). Natural gas may serve as a steppingstone to help transition from coal to cleaner energy sources, but it still produces large amounts of CO_2 and must not dominate the market if the grid is to be decarbonized. The output of fracking wells drops by up to 70% after the first year of production, which may slow the fracking boom and make natural gas still more competitive than coal but less competitive than low-carbon energy sources (Andrews & Jelley, 2022, p. 90).

Hydropower is another foundational energy source of the 20th century. It benefits from having low carbon emissions but faces challenges expanding in the 21st century. In 1985, hydropower provided 20% of the world's electricity; however, that number has dropped to 15% in 2022, and hydroelectricity generation in the US has flatlined since the 1970's (Saundry, Hydroelectricity in Context, 2024). Hydropower capacity is largely limited by geography, and in the US, if all new possible dams were built, it would not even double the existing capacity (Saundry, Hydropower Policy, 2024). High upfront capital costs and pushback against environmental impacts also limit new hydropower in developed nations (Saundry, Hydroelectricity in Context, 2024).

In developing nations, the potential for expanding hydroelectric power is much larger due to fewer regulations and more untapped opportunities to build dams. The International

Energy Agency (IEA) expects that Africa, Latin America, and Asia, will reach generation levels of 350 TWh, 1600 TWh, and 3000 TWh respectively by 2050 (Andrews & Jelley, 2022, p. 200). Where hydropower can be built, it has the potential to replace some percent of fossil fuel use as a steady, reliable, and easily ramped up power source, but in the developed world, outfitting existing dams for hydropower may be the most cost-effective way to increase hydropower output by 10-15% (Saundry, Hydroelectricity in Context, 2024).

Like hydropower, nuclear electricity generation thrived in the 20th century, accounting for 17.5% of global electricity generation in the 1990's. It has struggled to stay competitive in the 21st century due to safety concerns and extremely high capital costs (Saundry, Nuclear Power in Context, 2024). Nuclear energy, like fossil fuels, relies on fuels that produce heat which can be then transformed to kinetic and then electric energy (Andrews & Jelley, 2022, p. 322-323). Unlike fossil fuels, nuclear energy is a low carbon energy source that with 93.1% efficiency, giving it massive potential to decarbonize the grid (Saundry, Nuclear in Context, 2024). Germany's energy policy to phase out nuclear before carbon illustrates the impact nuclear can have on carbon emissions. While nearly 50% of Germany's electricity comes from renewables, its carbon emissions are nearly 8 times those of France. France gets only 30% of its electricity from renewables but gets two thirds of its electricity from nuclear energy (Saundry, Nuclear Power in Context, 2024).

Current nuclear technologies face challenges due to their boutique nature, with custom designs, manufacturing, and regulatory guidance making construction costs prohibitively expensive. The high cost means that future demand needs to be almost guaranteed to justify the construction of new nuclear facilities (Saundry, Nuclear Economics, 2024). Recent demand from AI and data centers has led to greater investment in modular nuclear energy (Conover, 2024). Modular nuclear units have the potential make nuclear more economic by building factory-fabricated reactors that can have standard production processes and be shipped to the generation site, potentially revamping nuclear in the 21st century and offering a low carbon alternative to fossil fuels (Andrews & Jelley, 2022, p. 336).

While traditional electricity generation sources are still prominent in the 21st century, wind and solar generation are becoming increasingly cost competitive, and new forms of technology are continuously developing. As seen in Figure 2, wind and solar began taking off in the US in 2007 and have been expanding quickly (EIA, 2024). Rapid expansion of these technologies coupled with the maintenance or expansion of nuclear and hydroelectric power and the phasing out of fossil fuels have great potential to decarbonize the grid.

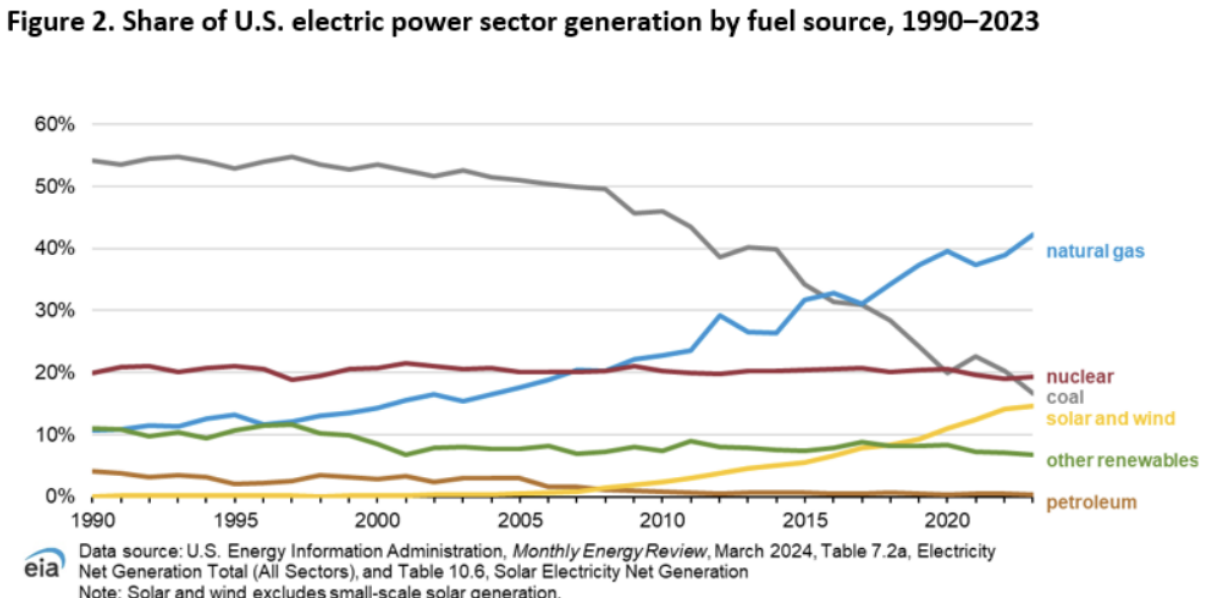

Figure 2

Wind energy offers vast potential for electricity generation with up to 560,000 TWh/y of onshore and 315,000 TWh/y of offshore globally (Andrews & Jelley, 2022, p.264). Wind power has technically been around for centuries but gone from a negligible contribution to US electricity generation at the turn of the century to reaching 10.2% of generation in the US by 2022 (Saundry, Wind Power in Context, 2024). Technological advancements, economies of scale, and policy support have led to a 63% reduction in LCOE from wind since 2009, making it increasingly competitive with more established forms of electricity generation (Saundry, Wind Power Economics, 2024).

There is still much engineering potential for designing more efficient turbines, but ultimately, wind power generation is incredibly dependent on wind speed and is capped at the Betz limit of ~59% efficiency (Andrews & Jelley, 2022, p. 232). Some of the greatest wind energy potential lies in offshore wind where wind speed tends to be high and turbulence tends to be low (Andrews & Jelley, 2022, p. 253). Cost effective technology for building offshore windfarms could thus accelerate wind production. Even so, wind speed can change quickly throughout the day, making wind a variable form of energy generation. Ideal locations for wind energy generation may also be far from areas of high population density. Thus, while wind generation may be cost effective, it requires infrastructure advancements to handle the variability in generation and transport it to areas of high demand.

Solar energy from photovoltaics has also taken off in the 21st century and suffers from a similar problem of variable generation. The LCOE for solar photovoltaic systems has

decreased 83% since 2009, making them more competitive with fossil fuels (Saundry, Policies and the Future of Solar, 2024). Globally, solar energy has a potential of at least 600,000 TWh per year, so as it becomes more cost effective, it has incredibly potential to reduce carbon emissions (Andrews & Jelley, 2022, p. 308). Solar voltaic energy benefits from the ability to be deployed at a variety of scales from small residential projects, to rural microgrids to large scale solar farms. In the U.S., 36% of solar projects are small scale distributed solar (Saundry, Solar Power in Context, 2024), and globally, 40% of capacity is distributed (Andrews & Jelley, 2022, p. 302).

The biggest drawback of solar electricity is the variability in generation. For the electric grid to be stable, supply must match demand at all times. However, solar energy supply peaks at midday and then decreases until nightfall while demand is often low during the day and spikes in the evenings in a phenomenon often called the duck curve (Andrews & Jelley, 2022, p. 367). Furthermore, it is sensitive to cloud coverage and changes with the seasons and the tilt of the Earth. To meet demand, solar energy must be accurately predicted on a daily basis and must be paired either with energy storage or fast ramping supplementary energy sources such as natural gas. Thus, while cheaper and more efficient photovoltaic cells could increase competitiveness, infrastructure changes and policy for integrating solar energy must take place, which often move slower than the pace of technological development.

Because of the challenges of integrating wind and solar energy into the existing electric grid, low carbon electricity generation on its own is not enough to decarbonize the global energy system. Technologies that assist in storing and transporting electricity are required as well as technologies that replace carbon intense fuel in transportation and heating. Existing electric grid infrastructure requires that supply always meet demand and works under the assumption that supply can be quickly ramped up when necessary to meet demand (Saundry, Electric Grid Overview, 2024). Solar and wind generation do not meet these criteria and therefore require innovations to overcome these challenges.

One potentially promising method of smoothing supply and demand is to expand electricity transmission using high voltage direct current (HVDC) lines instead of high voltage alternating current (HVAC) lines. HVDC power transmission is more efficient over long distances and underground than HVAC; it can have an efficiency of around 97% for a 1000km (Andrews & Jelley, 2022, p. 363). Being able to efficiently transport electricity long distances would allow for the transport of wind energy from areas of high wind velocity which tend to be remote to areas of high population density. It would also potentially allow for the transmission of wind and solar from areas with decorrelated weather patterns. This would mean that when there are clouds in one area of the country, energy can be supplied

from solar farms elsewhere with clear skies. Solar energy in the west at mid-day pacific time could also be used to support demand increases on the East Coast in the evening, helping to smooth out the duck curve (Andrews & Jelley, 2022, p. 366). Additionally, HVDC lines could help efficiently transport electricity from offshore wind farms to densely populated regions.

One challenge with HVDC transmission is that it requires converting the current from AC at generation to DC for transmission and back to AC for distribution since existing infrastructure uses AC (Saundry, Transformers, Transmission, and Distribution, 2024). Additionally, it requires the construction of new transmission lines which is very expensive and involves a great deal of inter-state bureaucracy or international politics (Saundry, Transformers, Transmission, and Distribution, 2024). There may be potential to refit existing transmission lines with HVDC when maintenance is inevitably required; however, since most existing lines are shorter to support HVAC, it is unlikely that doing so would allow for the interconnection of renewable energy generation from areas with uncorrelated weather patterns.

An HVDC interconnected grid may help reduce the variability of energy generation from wind and solar, but it would not be able to completely account for daily or seasonal trends. Today, wind and solar energy are often used to meet base demand and dispatchable generation from natural gas and other sources is ramped up or down as needed such as in the evening when solar energy output is low but demand is high. When wind and solar supply is high and demand is low, the energy goes unused and is called "curtailment" (Andrews & Jelley, 2022, p. 368). Energy storage technologies would allow excess energy generated when generation capacity is high and demand is low to be released later when generation capacity is low and demand is high.

Battery storage technologies have become dramatically more effective and affordable in the 21st century with the cost of lithium-ion batteries dropping from $1200 per kWh to $140 in a single decade and projected to continue dropping (Andrews & Jelley, 2022, p. 377). Modern lithium-ion batteries also have a high energy density, a long life time, and 80% efficiency, making them an attractive option for storing energy (Saundry, Batteries, 2024). Battery design and cost effectiveness is likely to improve in the future, but at present they suffer from physical limitations where many operate at very high temperatures or contain components that are unstable and difficult to source (Saundry, Batteries, 2024). An alternative to battery storage is pumped hydro storage which operates at about 80% efficiency and currently makes up 70% of the world's stored energy (Saundry, Pumped Hydroelectric Storage, 2024). Even so, it is largely underused with an estimated global seasonal storage potential of 17.3 PWh (Andrews & Jelley, 2022, p. 385). Expanding solar

and wind generation in conjunction with battery and pumped hydro storage could have a large impact on decarbonizing electricity.

Table 1. Total U.S. energy-related CO_2 emissions by sector, 2019–2023

million metric tons of carbon dioxide

Sector	2023	2022	2021	2020	2019
Residential	311	339	325	319	347
Commercial	250	261	245	233	255
Industrial	963	960	977	952	1,007
Transportation	1,856	1,840	1,807	1,630	1,921
Electric power	1,427	1,542	1,551	1,450	1,618
Total	**4,807**	**4,941**	**4,905**	**4,584**	**5,147**

Data source: U.S. Energy Information Administration, *Monthly Energy Review*, March 2024, Tables 11.1–11.6
Note: Totals may not equal sum of components due to independent rounding.

Table 1

As seen in Table 1, even with 100% decarbonizing electricity, almost 70% of carbon emissions in the US still come from other sectors like transportation as well as heating and cooling of industrial, commercial, and residential buildings (EIA, 2024). Once electricity has been decarbonized, then electrifying processes that currently use fossil fuels such as transportation would help to neutralize carbon emissions. In the US in 2024, 89% of energy for transportation came from fossil fuels, and transportation accounted for 30% of all energy consumption (Saundry, Transportation in Context, 2024).

Electric vehicles have grown in popularity in the 21st century as the cost of lithium-ion batteries has decreased. New car batteries have more mileage per charge, reducing "range anxiety" or the fear of being stuck somewhere without a charging station (Andrews & Jelley, 2022, p. 429). As more people buy electric vehicles, there will be more investment in infrastructure that makes it possible for more people to buy electric vehicles. An alternative to electrifying transportation is using renewable biofuels, however first-generation biofuels compete for food production and may contribute to global warming by replacing forested areas with farmland (Andrews & Jelley, 2022, p. 147). Electric vehicles seem much better poised to replace fossil fuels than biofuels at present, though this may change.

Heating and cooling of residential, commercial, and industrial buildings represent another primary energy use, drawing on both electricity and fossil fuels. Carbon intensity of heating could be reduced through improving the efficiency of heating and cooling systems or

leveraging geothermal heat pumps. Setting building codes that require new buildings to have high quality insulation and replacing windows and wall insulation in existing buildings could reduce energy consumption by 50-75% (Andrews & Jelley, 2022, p. 408). Geothermal heat pumps use the consistency of underground temperatures to warm or cool buildings, and IRENA estimates heat pumps could supply 4000 TWh per year of energy by 2050 (Andrews & Jelley, 2022, p. 179). For industrial heat needs such as steel manufacturing, electrification of heat could potentially replace fossil fuel burning; however, with current technology, decarbonizing industrial heating would increase electricity demand by 50% and result in a rise in costs between 20-100% (Andrews & Jelley, 2022, p. 418). More research and development are likely needed to achieve desired emission reductions for heat intensive industrial processes.

The technologies previously discussed are mature enough and largely cost effective enough to be widely implemented, but it took decades of research and development to get there. Many promising technologies remain under research today, with the potential to transform the global energy system in the coming decades. Wave and tidal hydropower, for instance, could convert ocean motion into electricity, potentially generating 400 TWh and 500 TWh annually, respectively (Andrews & Jelley, 2022, p. 210). However, these technologies currently face significant environmental and engineering challenges that limit their competitiveness. Floating wind farms represent another emerging technology, with the first operational farm beginning in 2017 and achieving a capacity factor of over 60% (Andrews & Jelley, 2022, p. 255). While still developing, these farms show promising potential for future energy generation.

There are many other technologies under development that could have the potential to help decarbonize the grid, but three technologies that have received some of the most mainstream attention are hydrogen, biofuels from algae, and nuclear fusion. Hydrogen can be thought of as an energy storage fuel because it takes energy to produce but can be burned at high heat with little emissions (Saundry, Fuel Cells and Hydrogen, 2024). Hydrogen could be produced by surplus renewable energy and then transported and used to provide heat and energy elsewhere. Unfortunately, hydrogen is not easy to transport, and the adoption of the necessary infrastructure to make it effective has proven slow and uncompetitive. Algae also has a great potential for use as fuel because they contain 20-50% of their mass in oil and can grow quickly in a small, unused area (Andrews & Jelley, 144). Mass production has proved challenging due to the risk of contamination and difficulty farming them (Saundry, Other Forms of Biomass Energy, 2024). Nuclear fusion has been a research focus since the 1960's and in recent years has made some progress towards feasible commercial implementation by 2050, but still requires considerable research (Andrews & Jelley, 2022, p. 350).

The transition to a decarbonized energy system represents a complex technological challenge that requires a multifaceted approach. Each energy technology discussed presents unique advantages and limitations, and different countries have different capacity to implement them. Successful decarbonization will depend on strategic integration of various technologies, considering factors such as regional resource availability, existing infrastructure, economic constraints, and grid stability. Continued research and development, along with adaptive implementation strategies, will be critical in addressing the technical challenges of transitioning away from fossil fuel-dependent energy systems while meeting growing global energy demands.

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